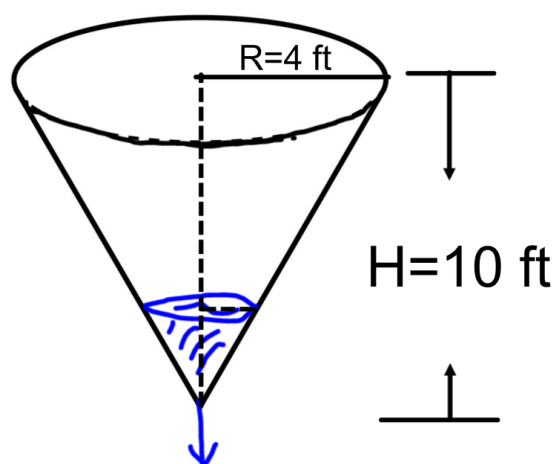


Problem 5

Problem 5

Problem 1 A conical water tank of height 10 feet and radius 4 feet is being drained through a hole in the bottom (See figure below).



- Let r be the radius of the water at depth h . In the figure above, label r and h .
- Find an expression for r in terms of h .
- The water in the tank is being drained at a rate of 2 feet^3 per minute. What is the rate of change of the water depth when the water depth is 3 feet?